Automatic multilabel electrocardiogram diagnosis of heart rhythm or conduction abnormalities with deep learning: a cohort study

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Summary

Background Market-applicable concurrent electrocardiogram (ECG) diagnosis for multiple heart abnormalities that covers a wide range of arrhythmias, with better-than-human accuracy, has not yet been developed. We therefore aimed to engineer a deep learning approach for the automated multilabel diagnosis of heart rhythm or conduction abnormalities by real-time ECG analysis.

Methods We used a dataset of ECGs (standard 10 s, 12-channel format) from adult patients (aged ≥18 years), with 21 distinct rhythm classes, including most types of heart rhythm or conduction abnormalities, for the diagnosis of arrhythmias at multilabel level. The ECGs were collected from three campuses of Tongji Hospital (Huazhong University of Science and Technology, Wuhan, China) and annotated by cardiologists. We used these datasets to develop a convolutional neural network approach to generate diagnoses of arrythmias. We collected a test dataset of ECGs from a new group of patients not included in the training dataset. The test dataset was annotated by consensus of a committee of board-certified, actively practicing cardiologists. To evaluate the performance of the model we assessed the F1 score and the area under the curve (AUC) of the receiver operating characteristic (ROC) curve, as well as quantifying sensitivity and specificity. To validate our results, findings for the test dataset were compared with diagnoses made by 53 ECG physicians working in cardiology departments who had a wide range of experience in ECG interpretation (range 0 to >12 years). An external public validation dataset of 962 ECGs from other hospitals was used to study generalisability of the diagnostic model.

Findings Our training and validation dataset comprised 180 112 ECGs from 70 692 patients, collected between Jan 1, 2012, and Apr 30, 2019. The test dataset comprised 828 ECGs corresponding to 828 new patients, recorded between Sept 11, 2012, and Aug 30, 2019. At the multilabel level, our deep learning approach to diagnosing heart abnormalities resulted in an exact match in 658 (80%) of 828 ECGs, exceeding the mean performance of physicians (552 [67%] for physicians with 0–6 years of experience; 571 [69%] for physicians with 7–12 years of experience; 621 [75%] for physicians with more than 12 years of experience). Our model had an overall mean F1 score of 0·887 compared with 0·789 for physicians with 0–6 years of experience, 0·815 for physicians with 7–12 years of experience, and 0·831 for physicians with more than 12 years of experience. The model had a mean AUC ROC score of 0·983 (95% CI 0·980–0·986), sensitivity of 0·867 (0·849–0·885) and specificity of 0·995 (0·994–0·996). Promising F1 scores were also obtained from the external public database using our proposed model without any model modifications (mean F1 scores of 0·845 in multilabel and 0·852 in single-label ECGs).

Interpretation Our model is more accurate than physicians working in cardiology departments at distinguishing a range of distinct arrhythmias in single-label and multilabel ECGs, laying a promising foundation for computational decision-support systems in clinical applications.

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Introduction

Electrocardiograms (ECGs) are a fundamental and reliable test for the diagnosis of heart arrhythmias.[1] Patients with multiple and concurrent arrhythmias diagnosed by ECG are commonly seen in clinical practice. Incorrectly interpreted ECGs might result in inappropriate clinical decisions or lead to adverse outcomes.[2,3]

A typical cardiologist will have upwards of 12 years of training, including undergraduate study, medical school, an internal medicine residency, and a cardiology fellowship. However, even for an experienced cardiologist, accurately interpreting the ECG for a patient with concurrent arrhythmias is challenging. Worldwide, more than 300 million ECGs are done annually, and in low-income and middle-income

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Research in context

Evidence before this study

We did a comprehensive literature survey of PubMed, Google Scholar, and PhysioNet for research articles and reviews on artificial intelligence-enabled diagnosis of electrocardiogram (ECG) arrhythmias published in English from March 12, 2016, to Feb 20, 2020, using the key words "deep learning" or "deep neural networks" or "convolutional neural networks" or artificial intelligence" and "electrocardiogram" or "arrhythmias" or "ECG" or "rhythm abnormality" or "conduction abnormality", with a special focus on articles with models that were developed using deep neural networks. We found that most of the previous deep learning models were trained and validated on benchmark datasets (such as the Massachusetts Institute of Technology-Beth Israel Hospital arrhythmia database, or the PhysioNet database), in which there were fewer than 15 disease classes or the included patients had only a single ECG abnormality, raising concerns about the generalisability and robustness of these algorithms. To the best of our knowledge, no study covers a wide range of rhythm or conduction abnormalities, and simultaneously uses deep learning algorithms to diagnose each abnormality with a high degree of accuracy. We identified two publications from Mayo Clinic in 2019, in which artificial intelligence-enabled diagnosis algorithms were developed and a classification of automatic atrial fibrillation or cardiac contractile dysfunction was achieved by using relevant, large datasets. We also identified another publication that reported that a deep learning approach can achieve a cardiologist-level performance in ECG arrhythmia detection for 12 rhythm classes, with the single-channel ECG format.

Added value of this study

This work focuses on patients with multiple and concurrent ECG arrhythmias in real clinical practice. Our constructed dataset retains the original format of the ECG results without further processing, and the annotation results are stored according to the initial diagnoses provided. We developed a convolutional neural network approach that yields correct interpretation of normal ECGs and a highly accurate multilabel diagnosis of 20 ECG arrhythmias, including all of the common rhythm or conduction abnormality types. Standard 12-channel, 10 s ECG signals were used directly from the three types of monitoring machines, negating the effort that otherwise would be needed for feature extraction processes. We assessed the performance of the deep learning model on an individual test dataset that was annotated by a consensus committee of board-certified, practising cardiologists. Performance of the model was compared with 53 physicians working in cardiology departments across a wide range of experience levels from 0 to more than 12 years. We found that our model exceeds the performance of physicians clinically trained in ECG interpretation. Furthermore, we simplified our dataset by separating the ECGs with multiple labels (patients with several diseases) and trained another deep learning model for multiclass ECG diagnosis. This enabled us to compare our algorithm with an existing model. Diagnosis results showed that our model is superior to the existing model.

Implications of all the available evidence

The accuracy of our deep learning algorithms and the wide coverage of our dataset support our hypothesis that artificial intelligence-enabled algorithms can match or exceed experts for automatically diagnosing ECGs of patients with rhythm or conduction abnormalities. The proposed algorithm could be embedded into ECG machines and deployed at the clinical level to assist the decision making process of electrocardiographers and physicians. This approach could help provide a cost-effective and efficient way for higher throughput, yet still highly accurate ECG interpretation. Our algorithm shows an ability to classify a wide range of distinct arrhythmias, under a single label or multiple labels, better than physicians, laying the foundation for the deployment of computational decision support systems in clinical practice.

countries, providing the correct diagnosis can be especially difficult because experienced cardiologists are scarce.[4] Consequently, there is an urgent unmet need for an automated or assisted multilabel ECG diagnostic framework, supported by artificial intelligence technology.[5]

During the past 2 years, several investigations have aimed to identify arrhythmias using deep learning techniques,[6–9] and several benchmark datasets (eg, the Massachusetts Institute of Technology-Beth Israel Hospital arrhythmia database,[10] American Heart Association database, and the Physionet database[11]) have been shown to be valuable for evaluating arrhythmia deep learning models on the basis of objective performance measures. Reports have shown that by using a robustly designed deep learning model, it is possible to identify specific abnormalities, such as ventricular premature contraction and atrial fibrillation.[12–14] However, most of studies that have been done so far have focused on ECG signal fragments that each contain a small number of labelled ECG samples and few types of arrhythmia, which makes it challenging to build an arrhythmia classifier that is competitive with human cardiologists.[15] Moreover, existing algorithms only use data from single-channel ECGs or single-originated ECGs (eg, from GE-Marquette type 3500 or 5500 machines),[6] yet there is an abundance of unused information in standard format raw data (without preprocessing) and in ECG samples from other diverse sources. The rational incorporation of these data could increase the applicability and the generalisability of the proposed machine learning algorithms.

By investigating the ECG diagnosis problem at a multilabel level, we aimed to address the problem associated with patients with more than one cardiac disease, which are frequently reported simultaneously during the clinical diagnosis process. We aimed to develop a general method, using ECG tests, for the end-to-end automatic diagnosis of heart conditions in patients, and to produce an artificial intelligence-based multilabel ECG diagnosis model that covers almost all types of arrhythmias caused by rhythm and conduction dysfunction incorporation with heart rate of the heart's electrical conduction system.

Methods

Data sources and study population

Our dataset comprised retrospective data from adult patients (aged ≥18 years) who had an ECG at three campuses (Main Campus, Optical Valley Campus, and Sino-French New City Campus) of the Cardiac Function Examination Center (Division of Cardiology, Tongji Hospital, Huazhong University of Science and Technology, Wuhan, China). The dataset included standard 10 s, 12-channel ECGs recorded at a sampling rate of 500 Hz using GE-Marquette type 3500 or 5500 ECG machines (GE Healthcare, Milwaukee, WI, USA), and 24 h dynamic 12-channel ECGs recorded by a Holter machine (DMS Holter Company, Stateline, NV, USA) or electrophysiological monitor (Dong Fang Company, Suzhou, China, only offering ECG data for paroxysmal supraventricular tachycardia). The raw data were stored using the MUSE data management system (GE Healthcare). The ECGs were classified into 21 specific rhythms: normal, sinus tachycardia, sinus bradycardia, premature atrial contraction, atrial rhythm, atrial tachycardia, atrial flutter, atrial fibrillation, premature junctional contraction, junctional rhythm, paroxysmal supraventricular tachycardia, premature ventricular contraction, idioventricular rhythm, ventricular tachycardia, artificial atrial pacing rhythm, artificial ventricular pacing rhythm, left bundle branch block, first-degree atrioventricular block, Mobitz type I second-degree atrioventricular block, Wolff-Parkinson-White syndrome type A, and Wolff-Parkinson-White syndrome type B. All the ECGs from this dataset were randomly assigned by using the shuffle method in Python to randomly order the ECGs, then assigning into training and validation sets with a ratio of 9:1, then removing repeated patients from each set so there was no overlap between the groups (figure 1).

An independent, open access dataset containing 6878 12-channel ECG abnormalities from 11 different hospitals was provided by the China Physiological Signal Challenge (launched Jan 15, 2018)[16] for use for external validation to examine the generalisability of the proposed approach. This external dataset included ECGs recorded for periods other than 10 s and included different abnormalities to our training dataset. To ensure the data format was consistent with the model input, we selected ECGs from the external dataset that had been recorded for longer than 10 s and cropped them to 10 s; of these cropped ECGs, we selected 962 with either normal rhythm or one of five of the same abnormalities as found in our dataset to use for the external validation (appendix 1 p 16).

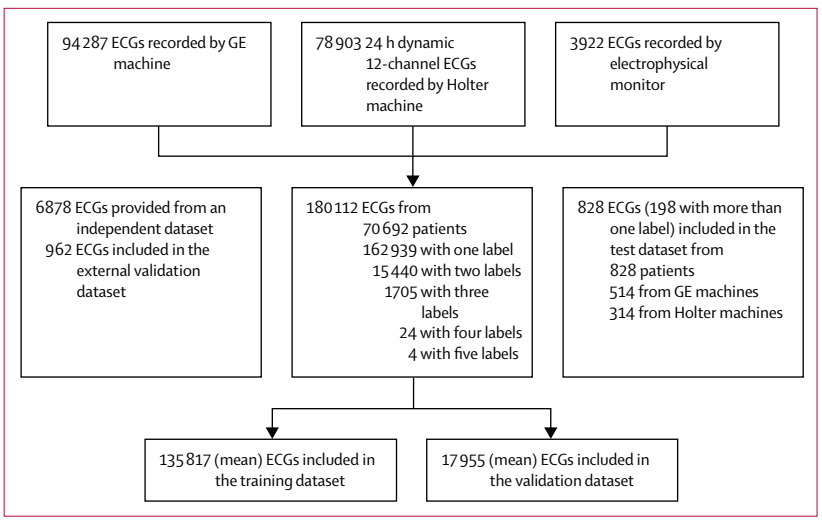


Figure 1: **Profile of the datasets**
ECGs in the training and validation dataset were recorded from Jan 1, 2012, to April 30, 2019. The ECGs in the test dataset were recorded between Sept 11, 2012, and Aug 30, 2019. To prevent time series dependencies of the datasets, we ensured that data from any individual patient was included either in the training, validation, or testing set. Each patient in the test dataset only contributed one ECG to that set. For the less common abnormalities, some patients contributed more than one ECG recording to ensure a balanced dataset for training. Random shuffle was applied to randomly assign the training and validation sets; for patients with more than one ECG recorded, their ECGs were only used for either training or validation to prevent patient overlapping. ECG=electrocardiogram. GE machine=GE-Marquette type 3500 or 5500 ECG machine.

See Online for appendix 1

To ensure patient confidentiality, we used anonymised data. The data collection team was responsible for sample collection and anonymisation, and the algorithm development team received anonymised data with only patient identification number, age, and sex information for the subsequent algorithm development. Written informed consent was not required for this study, because the ECG samples were appropriately anonymised and de-identified, as per the Health Insurance Portability and Accountability Act Safe Harbor provision.[17] The study design was evaluated and exempted from full review by the Huazhong University of Science and Technology Institutional Review Board.

Annotation procedure

ECGs from the training and validation datasets were initially interpreted by a primary cardiologist and then further reviewed by senior board-certified cardiologists who were actively practising and working in the Cardiac Function Examination Center of Tongji Hospital, to ensure the correctness of the base diagnostic labels. Labels of the 12-channel ECGs are consistent with the diagnostic results of clinical cardiologists. The procedure

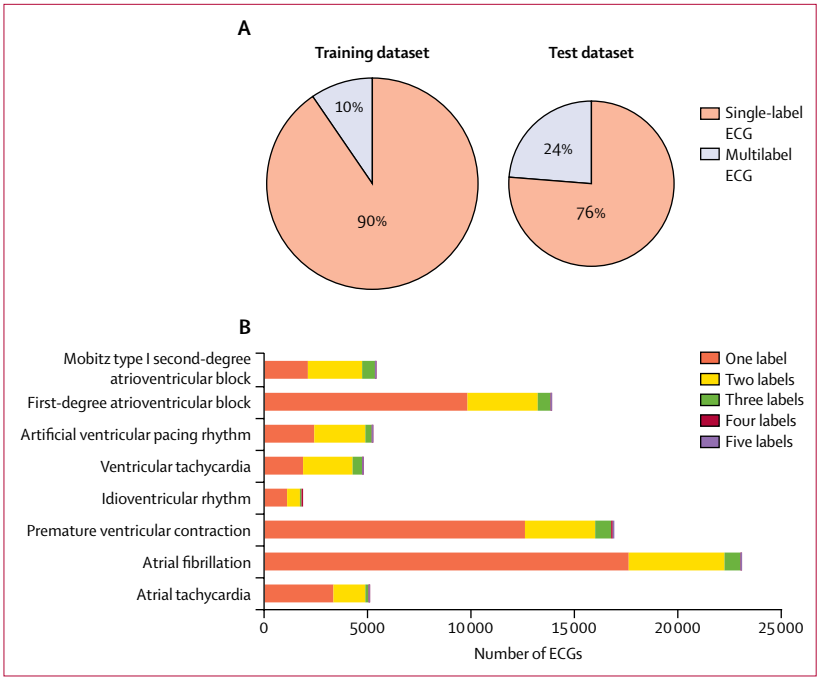


Figure 2: **Proportion of multilabel ECGs in the constructed dataset**
(A) Proportion of multilabel ECGs in the training and test datasets. (B) Proportion of multilabel ECGs within eight of the included rhythm classes. ECG=electrocardiogram.

for segmenting the ECGs from the Holter machine is shown in appendix 1 (p 1).

We collected another set of ECG results to construct a test dataset corresponding to a new group of patients, who were not included in the training dataset. A committee with three board-certified actively practising cardiologists, including one cardiac electrophysiologist, annotated ECGs in this test dataset. Cardiologist committees discussed the records, annotated by consensus, and removed ECGs about which they did not reach an agreement, providing an expert standard for model evaluation. ECGs that could not be interpreted or agreed by the committee were removed from our test dataset.

Physician evaluation

To help assess the comparative performance of the deep learning model, 53 ECG physicians working in cardiology departments contributed directly to this study. All participating physicians received specific instructions and training regarding how to annotate ECGs to ensure consistency. We divided the physicians into three groups according to the time they had been working with ECGs: 0–6 years, 7–12 years, and more than 12 years. Appendix 1 (p 4) shows the number of physicians, as well as the mean age, experience level, and sex proportion, of each group. Each group received the whole test dataset and ECG physicians in each group reviewed a randomly selected (using a random number table) set of the ECGs in a non-repetitive manner.

Overview of the artificial intelligence model

We developed a convolutional neural network approach. The standard 10 s, 12-channel raw ECG (sampled at 500 Hz) was transformed to a 5000×12 matrix, which was used as the input for the convolutional neural network model. The output of the model is a 1×21 vector, and each element of this vector corresponds to one specific ECG rhythm class. Output vectors of ECGs from patients who have multiple arrhythmias contain more than one non-zero element.

The proposed convolutional neural network model has 15 alternating layers (appendix 1 pp 4–6) for the purpose of multilabel classification of the 21 rhythm types. To prevent gradient vanishing problems, shortcut connections[18] in residual blocks were used to skip intermediate layers. Rectified linear unit non-linearity with dropout[19] was used in the network to provide a better model convergence rate and prevent overfitting. The model was trained using the multiclass cross-entropy loss function, in which the weight of a class is the ratio of the maximum sample quantity of all classes relative to the sample quantity of that class. The output layer generates the diagnosis results with 21 sigmoid functions that return distributions corresponding to the 21 output classes. Network structure, hyperparameters, such as filter number, kernel size and max-pooling size of each layer, are shown in appendix 1 (pp 4–6).

Statistical analysis

The performance of the model was evaluated by assessing its prediction accuracy, the area under the curve (AUC) of the receiver operating characteristic (ROC), sensitivity, specificity, and F1 score (harmonic mean of the predictive positive value and sensitivity), with two-sided 95% CIs. Confusion matrices were used to evaluate whether predictions were discordant with the committee consensus. For the committee's annotations in the test dataset, interannotator agreement was calculated as the ratio of the number of times three annotators agreed that a rhythm was present to the total number of all ECGs in the test dataset.

For the prediction accuracy, the concept subset accuracy[20] (ie, exact match accuracy) was used to evaluate the ratio between the predicted label

$$\hat{Y}_s$$

and the ground-truth label Y_s, according to the following expression:

$$\text{Accuracy}=\frac{1}{S}\sum_{s=1}^{S}1_{\{\hat{Y}_s=Y_s\}}$$

where the sum is of the indicator function and S is the ECG sample number to be assessed.

Precision, sensitivity, specificity, and F1 score of each class rhythm n, where n is a positive integer, are related

	Model AUC ROC (95% CI)	Model sensitivity (95% CI)	Model specificity (95% CI)	Model F1 score (95% CI)	Physicians' mean F1 score
Normal	1·000 (1·000–1·000)	1·000 (1·000–1·000)	1·000 (0·999–1·000)	0·998 (0·994–1·000)	0·834
Atrial flutter	0·993 (0·99–0·995)	0·898 (0·884–0·912)	0·992 (0·991–0·993)	0·880 (0·867–0·893)	0·836
Atrial fibrillation	0·991 (0·989–0·993)	0·873 (0·856–0·889)	0·985 (0·982–0·987)	0·863 (0·848–0·877)	0·839
Paroxysmal supraventricular tachycardia	0·982 (0·976–0·987)	0·895 (0·887–0·904)	0·999 (0·998–0·999)	0·931 (0·927–0·936)	0·808
Artificial atrial pacing rhythm	1·000 (0·999–1·000)	0·986 (0·972–1·000)	1·000 (1·000–1·000)	0·992 (0·983–1·000)	0·947
Artificial ventricular pacing rhythm	0·990 (0·988–0·991)	0·890 (0·880–0·900)	0·997 (0·996–0·997)	0·917 (0·909–0·924)	0·909
Mobitz type I second-degree atrioventricular block	1·000 (1·000–1·000)	0·988 (0·976–0·999)	1·000 (0·999–1·000)	0·989 (0·981–0·997)	0·952
Wolff-Parkinson-White syndrome type B	0·997 (0·995–0·999)	0·917 (0·899–0·935)	0·998 (0·997–0·999)	0·941 (0·931–0·951)	0·858
Wolff-Parkinson-White syndrome type A	1·000 (1·000–1·000)	0·991 (0·983–0·998)	0·997 (0·996–0·997)	0·972 (0·967–0·978)	0·891
Mean	0·995 (0·993–0·996)	0·937 (0·926–0·948)	0·996 (0·995–0·997)	0·943 (0·934–0·951)	0·875

Nine of the most common arrhythmias are shown here; results for other classes are shown in appendix 1 (p 9). AUC=area under the curve. ROC=receiver operating characteristic.

Table: **Performance summary of the deep learning model in arrhythmia diagnosis, and F1 scores for the deep learning model and physicians working in cardiology departments**

to true-positive (TP), true-negative (TN), false-positive (FP), and false-negative (FN) rates,[21] with formulas as follows:

$$\text{precision}_n = \frac{TP}{TP+FP}$$

$$\text{sensitivity}_n = \frac{TP}{TP+FN}$$

$$\text{specificity}_n = \frac{TN}{TN+FP}$$

$$F1_n = \frac{2 \times \text{precision}_n \times \text{sensitivity}_n}{\text{precision}_n + \text{sensitivity}_n}$$

Macro mean F1 score is then employed to evaluate the overall performance across all classes of data:

$$\text{Mean F1} = \frac{1}{N}\sum_{n=1}^{N} F1_n$$

Our algorithm was compared with another artificial intelligence-assisted diagnosis model with multiclass single-label ECGs, proposed by Hannun and colleagues.[17] We could only compare performance at the multiclass single-label level (instead of the multilabel level used in this study) because the model by Hannun and colleagues[17] was mostly trained on single-label ECG data. When ECGs with multiple labels were removed from the training, validation, and test datasets, 162 939 ECGs were used for training and 630 for testing the single-label model.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

Our dataset consisted of 180 112 ECGs from 70 692 adult patients, recorded for 10 s, using 12-channel ECG machines, between Jan 1, 2012, and Apr 30, 2019. The ratio of patients tested at each of the three campuses (ratio of Main Campus to Optical Valley Campus to Sino-French New City Campus) was approximately 3·7 to 2·6 to 1·1. There were at least 4000 samples showing each abnormality, except idioventricular rhythm (1780 samples because of the rarity of this disease), allowing the compilation of a balanced dataset. The mean age of the patients was 57·9 years (SD 15·8; appendix 1 p 1) and 32 377 (45·8%) were women. For some severe malignant arrhythmias, such as ventricular tachycardia, atrial fibrillation, and atrial tachycardia, more than 50% of the ECGs were from patients who were admitted to hospital after clinical events, whereas more than 99% of normal results, sinus brachycardia, and sinus tachycardia were recorded when individuals attended primary health-care services. We validated our deep learning model with the test dataset consisting of 828 ECGs recorded between May 1, 2019, and Aug 30, 2019, corresponding to 828 unique patients (488 [59%] women; mean age 50·8 years [SD 17·1]). For the diagnoses provided by the committee for the test dataset, inter-annotator agreement occurred in 745 (90%) of cases.

17 199 (10%) of the 180 112 ECGs in the training dataset were multilabel, as were 198 (24%) of 828 in the test dataset (figure 2A). In Mobitz type I second-degree atrioventricular block, artificial ventricular pacing rhythm, and ventricular tachycardia, the number of ECGs with more than one label was similar to the number of single-label ECGs, and for each disease shown in figure 2B there were ECGs with up to five labels. Detailed information about the distributions of labels is shown in appendix 1 (pp 3–4).

We chose the deep learning model with the lowest error in the validation dataset as the final model for

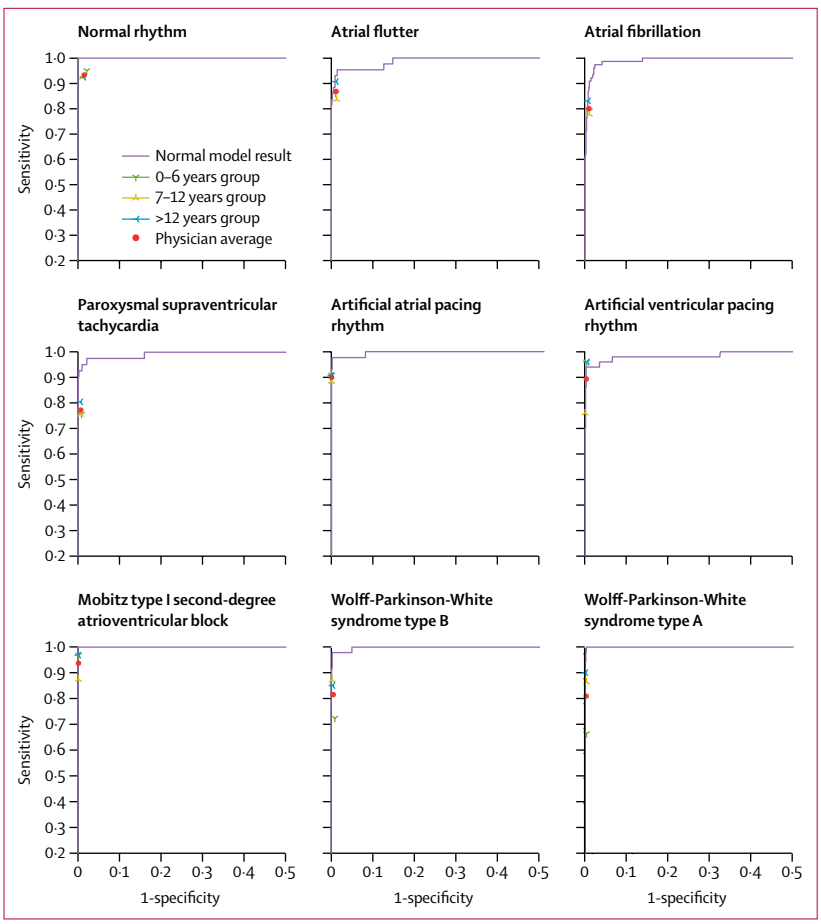


Figure 3: **ROC curves of prediction sensitivity of the deep learning model for nine of the included rhythm classes, compared with physicians**
ROC=receiver operating characteristic.

See Online for appendices 2–5

further investigation. The deep learning model was accurate in 658 (80%) of 828 cases in the test dataset, which was better than the 70% mean accuracy of physicians (552 [67%] for physicians with 0–6 years of experience, 571 [69%] for 7–12 years, and 621 [75%] for more than 12 years). Appendices 2–5 show detailed prediction results of each ECG of each class rhythm.

We compared the F1 scores of the model with those of the ECG physicians for nine most common rhythm classes (table). Using the committee labels as the expert standards, F1 scores of the physicians increased as the number of years working in ECG interpretation increased (appendix 1 p 10). The mean F1 score of our algorithm for the representative nine arrhythmias was 0·943 (95% CI 0·934–0·951), which exceeded the mean score 0·875 of the physicians. The F1 scores obtained by the model for the nine representative rhythm classes (table) were all higher than the average scores of the physicians in different rhythm sites incorporating different heart rates, including pacemaker and conduction types. Across all 21 classifications, our model had an overall mean F1 score of 0·887 compared with 0·789 for physicians with 0–6 years of experience, 0·815 for physicians with 7–12 years of experience, and 0·831 for physicians with more than 12 years of experience (appendix 1 p 9). The model had F1 scores ranging from 0·692 (sinus brachycardia) to 0·998 (normal); in 19 out of the 21 rhythm classes, the algorithm, on average, had higher F1 scores than the physicians.

The mean AUC ROC score of the algorithm for diagnosing all 21 classifications is 0·983 (95% CI 0·980–0·986), with sensitivity 0·867 (0·849–0·885) and specificity 0·995 (0·994–0·996; appendix 1 p 8). The lowest AUC ROC score was 0·901 (95% CI 0·896–0·906) for sinus brachycardia and the highest was 1·000 (1·000–1·000) for normal, artificial atrial pacing rhythm, Mobitz type I second-degree atrioventricular block, and Wolff-Parkinson-White syndrome type A. Depending on the impulse site, the class-weighted mean AUC ROC were 0·957 for the detection of sinus (normal, sinus tachycardia, sinus brachycardia), 0·983 for atrial (premature atrial contraction, atrial rhythm, atrial tachycardia, atrial fibrillation, atrial flutter), 0·982 for junctional (premature junctional contraction, junctional rhythm, paroxysmal supraventricular tachycardia), 0·983 for ventricular (premature ventricular contraction, idioventricular rhythm, ventricular tachycardia), and 0·995 for artificial pacemaker (artificial atrial pacing rhythm, artificial ventricular pacing rhythm) classes. The class-weighted mean AUC ROC was 0·982 for assessing tachycardia (sinus tachycardia, atrial tachycardia, paroxysmal supraventricular tachycardia, ventricular tachycardia) and 0·967 for bradycardia (sinus brachycardia, atrial rhythm, junctional rhythm, idioventricular rhythm). For diagnosis of the conduction-type abnormalities (left bundle branch block, first-degree atrioventricular block, Mobitz type I second-degree atrioventricular block, Wolff-Parkinson-White syndrome type A and B), the mean AUC ROC was 0·994.

When comparing ROC curves of the model with those of the physicians (mean across all physician groups), the model performed better, except in diagnosing premature ventricular contraction and ventricular tachycardia (figure 3; appendix 1 pp 10–11). The performance of our model was slightly worse than physicians with more than 12 years of experience for diagnosing sinus brachycardia, premature ventricular contraction, ventricular tachycardia, and artificial ventricular pacing rhythm, but better than the group with 7–12 years of experience for most diagnoses except ventricular tachycardia. The confusion matrices of the model show higher or similar TP rates compared with the physicians (figure 4; appendix 1 pp 12–13). The model also shows much lower FN rates than the mean for the ECG physicians in all of the abnormalities except sinus brachycardia, premature atrial contraction, premature ventricular contraction, and ventricular tachycardia.

In the test dataset, there were 37 atrial fibrillation rhythms with multiple labels and 40 atrial fibrillation

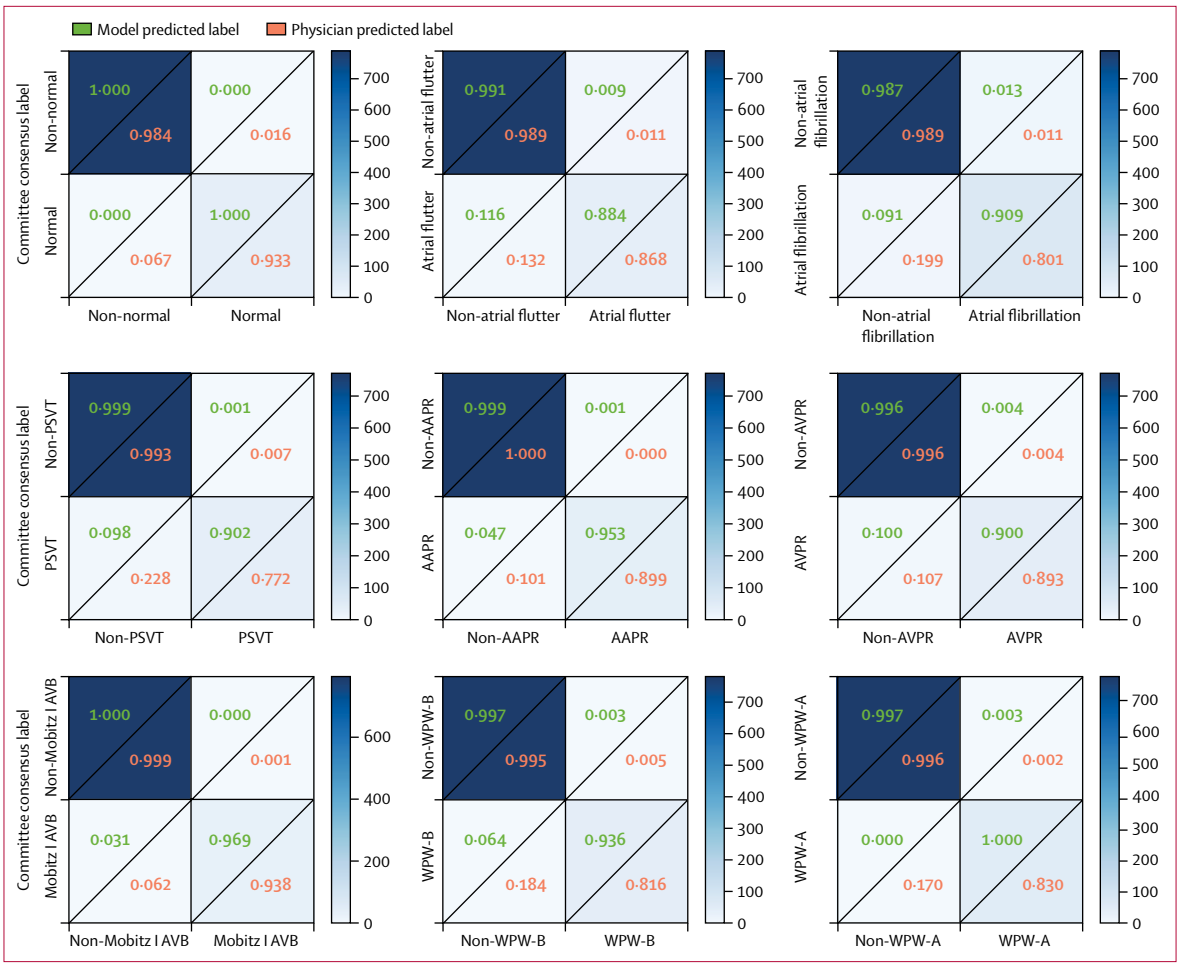


Figure 4: **Multilabel confusion matrices of the deep learning model and physicians**
AAPR=artificial atrial pacing rhythm. AVPR=artificial ventricular pacing rhythm. Mobitz I AVB=Mobitz type I second-degree atrioventricular block. PVST=paroxysmal supraventricular tachycardia. WPW-A=Wolff-Parkinson-White syndrome type A. WPW-B=Wolff-Parkinson-White syndrome type B.

rhythms with a single label. 29 of those multiple labels and 38 of the single labels were correctly diagnosed using the model, which is more than the average correct diagnoses made by the physicians, with 19·3 for multiple labels and 36·7 for single label (figure 5). For the multilabel ECGs, the model wrongly diagnosed three atrial fibrillation rhythms as atrial flutter; missed the diagnoses of three atrial fibrillation rhythms, two premature ventricular contractions, one ventricular tachycardia, one artificial ventricular pacing rhythm, and one Wolff-Parkinson-White syndrome type B; and wrongly diagnosed one artificial ventricular pacing rhythm and one Wolff-Parkinson-White syndrome type B in cases in which no abnormality was diagnosed by the consensus committee.

Results from the test dataset show that our model has higher F1 scores for most of the listed rhythms compared with the sequence level results from the model by Hannun and colleagues[17] (appendix 1 p 15). The model mean F1 score (0·916) is higher than the mean F1 score for physicians (0·801; appendix 1 p 15). TP values on the diagonal line in the model's confusion matrix are all higher than the corresponding values of ECG physicians (appendix 1 p 14). The two confusion matrices have a similar colour distribution for different class rhythms, showing that the model learns the equivalent hidden patterns and representative features recognised by physicians.

For the external published validation data, our model (without any model modifications) had mean F1 scores of 0·845 in multilabel ECGs and 0·852 in single-label ECGs (appendix 1 pp 16), which are similar to the results (mean of the normal rhythm and same five abnormal rhythms found in both datasets) from the test dataset constructed by patients from Tongji Hospital (0·025 lower than the mean F1 scores for the multilabel model, and 0·076 lower than the mean F1 scores for the single-label model). The above results demonstrate that the proposed algorithm appeared to be generalisable across different patients from different hospitals.

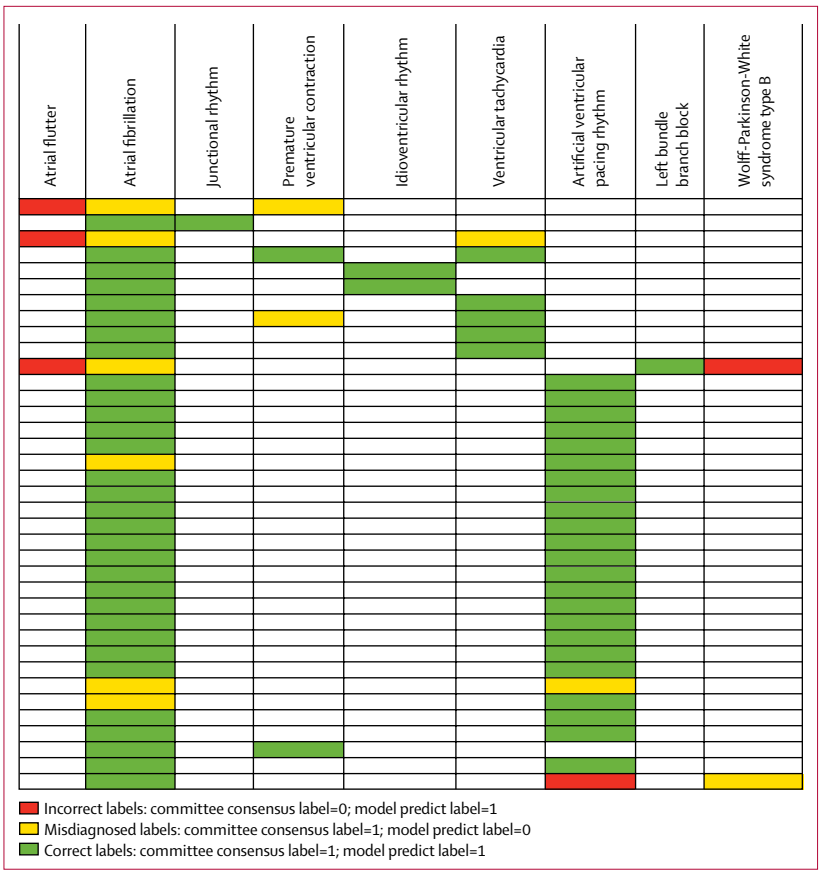


Figure 5: **Model atrial fibrillation predicted results compared with the committee consensus labels in the test dataset**

37 ECG results with more than one label of atrial fibrillation data are presented. ECG=electrocardiogram.

Discussion

Diagnosis of rhythm or conduction abnormalities is thought to be the most important factor in ensuring precise clinical decisions, and timely clinical treatment.[22] ECG provides valuable information about the presence, extent, and severity of arrhythmias such as atrial fibrillation, ventricular tachycardia, and atrioventricular block.[23,24] However, accurate diagnosis of rhythm or conduction abnormalities by experts presents various challenges. Most crucially, the precision of diagnosis by a physician might be insufficient for highly similar ECG waveforms. Several rhythm and conduction abnormalities (eg, sinus tachycardia and sinus brachycardia, atrial rhythm and atrial tachycardia, idioventricular rhythm and ventricular tachycardia) mainly differ in the heartbeat rhythms and are difficult to visually distinguish, particularly for doctors with little experience interpreting ECGs.[25] The artificial intelligence-based ECG diagnosis model developed in this study was trained using a large cohort dataset that included 20 types of abnormal ECGs and a single normal counterpart. Within the dataset, there was comprehensive representation of rhythm and conduction abnormalities, among which the rhythm classification consists of sinus, atrial, junctional, ventricular, and artificial pacemaker, and the conduction classification consists of bundle branch block, atrioventricular block and Wolff-Parkinson-White. The specific heart rate is also incorporated into these abnormalities, thus increasing the complexity of the dataset and the diversity of the abnormalities included. As well as the heart's inherent impulse site, we included the artificial pacemaker impulse site in the atrial and ventricular type. The model achieved higher accuracy and F1 scores in most of the 21 ECG types compared with the average scores of the ECG physicians, indicating that the model is a promising tool in artificial intelligence-enabled ECG diagnosis.

All the ECGs in our study were 12-channel type, which could provide essential information for accurately diagnosing some specific arrhythmias. For example, Wolff-Parkinson-White type A and B exhibit similar overall waveforms and just marginal differences in channel V1.[25] For diagnosis of the conduction-type abnormalities the mean AUC ROC of 0·994 indicated that our algorithm can successfully extract and distinguish complex features even for highly similar ECG waveforms such as left bundle branch block and Wolff-Parkinson-White syndrome type B. The model can detect features that might not be seen by a physician, resulting in improved diagnosis of Wolff-Parkinson-White type A and B, and higher F1 scores than those of ECG physicians. Additionally, ECGs in our study are from diverse sources, including conventional GE machines and dynamic ECG machines, and from a large group of patients, which included similar numbers of men and women with many distinct clinical heart conditions, and a range of ages and patients presenting with more than one arrhythmia, which also increased the diversity of the diseases represented in our dataset.

Previous studies generally used only single-channel ECG recordings,[26,27] whereas our study is innovative because we used multilabel classifications of the ECG data, which detect multiple abnormalities within a single period of monitoring in patients. For example, atrial fibrillation is one of the most common sustained cardiac arrhythmia,[28] often with comorbidities, which increase the difficulty of making a timely and accurate diagnosis. In our test dataset, atrial fibrillation ECGs were diagnosed with premature ventricular contraction, artificial ventricular pacing rhythm, idioventricular rhythm, ventricular tachycardia, left bundle branch block, and ventricular tachycardia plus premature ventricular contraction, which is close to what we would expect in a clinical setting (figure 5).

The results from our automatic ECG diagnosis system are similar to or better than those produced by physicians, as shown by applying our algorithm on a test dataset that was annotated by the consensus of a committee of board-certified actively practising cardiologists. We found that the model had much lower FN rates than ECG physicians for all abnormalities except sinus

brachycardia, premature atrial contraction, premature ventricular contraction, and ventricular tachycardia, demonstrating that the proposed model can efficiently prevent undertreatment of patients with rhythm and conduction abnormalities. Additionally, we developed a method to compare our algorithm with another single-label model.[17] An external public test dataset from 11 different hospitals was also used to show the generalisability of our proposed algorithm. These comparison results show that our model exhibits better results in terms of both the classification accuracies and the F1 scores of different class rhythms.

The deep learning model in this study outperformed experienced physicians working in cardiology departments. However, this study had some limitations. A few types of rhythm and conduction arrhythmias were not included because of an absence of patient samples. Also, ECG abnormalities such as myocardial infarction and atrial or ventricular hypertrophy have not been included, because these abnormalities tend to be characterised mainly by the overall shape of waveforms rather than by the characteristic rhythm and conduction dysfunctions as studied here. The model wrongly diagnosed or missed several diagnoses for multilabel ECGs: three atrial fibrillation rhythms were diagnosed as atrial flutter; the diagnoses of three atrial fibrillation rhythms, two premature ventricular contractions, one ventricular tachycardia, one artificial ventricular pacing rhythm, and one Wolff-Parkinson-White syndrome type B were missed; and one artificial ventricular pacing rhythm and one Wolff-Parkinson-White syndrome type B were diagnosed in patients with no arrhythmia diagnosed by the consensus committee. We think this is because complex arrhythmias result in irregular waveforms, thus increasing the difficulty of ECG diagnosis.

Moreover, some of the physicians with 0–6 years of experience had only recently completed their medical degree and obtained their physician certificate, which is likely to have reduced the overall score of that group. Additionally, experienced physicians who are accustomed to interpreting ECGs with the main label but ignoring concomitant diagnoses that are generally thought to be less important for patient treatment, could have reduced the overall score of the physicians. Even at the expert-level gold standard achieved by committee members, fallibility is inevitable with a very large sample size. Another limitation is that the patients included in this study were mainly from the central part of mainland China. Although variation in this population was well captured by our dataset, the demographic coverage is still lacking because of the limited geographical diversity of the patients.

To the best of our knowledge, this is the first time that a deep learning approach has been used to systematically study almost all types of arrhythmias caused by rhythm and conduction dysfunction, leading to an end-to-end, artificial intelligence-based, multilabel ECG diagnosis model. With advances in internet technology, we foresee this algorithm having a particularly promising use in telemedicine, especially for centres and hospitals (eg, in rural areas of China) where experienced cardiologists or technicians are scarce. Even now, the proposed algorithm could be directly implemented and deployed at the clinical level to assist the decision making process of electrocardiographers and physicians. The biggest advantages of our model are the cost-effectiveness and efficiency with respect to automated ECG interpretation. Our algorithm could potentially be used for real-time diagnosis through wearable devices that monitor heart and cardiovascular conditions via ECG signals. In principle, our algorithm shows a superior capability to classify a wide range of distinct arrhythmias, with single or multiple labels, with accuracy, precision, and efficiency, as compared with human experts, laying a foundation for the deployment of computational decision-support systems in real-world clinical environments.

Contributors

HZ and CC contributed equally to the article and wrote the manuscript. HZ, YX, BW, and XY analysed the data. CC, HY, XL, JD, BZ, and YY developed and coded the models. GW, XY, and YY designed the experiments and edited the manuscript. PZ and FL annotated the test dataset as two of the three board-certified cardiologists in the committee. JW, YL, SH, BW and YX collected the ECG data.

Declaration of interests

GW has received a grant from the National Natural Science Foundation of China, during the conduct of the study. XY has received grants from Hubei Science and the Technology Project of China, during the conduct of the study. HZ, CC, HY, XL, JW, XY, and YY have applied for a patent for the deep learning algorithm in this study, pending to Huazhong University of Science and Technology. All other authors declare no competing interests.

Data sharing

Python scripts for data preprocessing and modelling can be requested by contacting corresponding author YY. The test dataset from Tongji Hospital of this study is publicly available at Mendeley Data, V1, DOI:10.17632/6jd4rn2z9x.1. The Independent China Physiological Signal Challenge dataset is a public dataset available at: http://2018.icbeb.org/Challenge.html. The training and validation sets from Tongji Hospital collected for model training will be available after approval by the corresponding author Xiaoyun Yang. Any data use will be restricted to non-commercial research purposes. Readers can submit proposals for the training and validation sets to Xiaoyun Yang.

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